

Project Deliverable 1 – Project Proposal

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Section A

Question 1

1.1) Introduction

South Africa faces a spiralling growth in unemployment that can negatively impact economic welfare and social stability. As a result, community entrepreneurship is encouraged to create sustainable economic development with local enterprises (Thusi, X. & Ndlovu, S.G. 2025: 659).

These local enterprises consist of informal traders where they perform their operations through street vending. The e-commerce market is being rapidly adopted throughout South Africa and is estimated to reach a market value of \$7.9 billion in 2027. Therefore, informal traders want to expand their operations into the e-commerce market due to benefits such as streamlined payment transactions, reduce financial limitations and stronger consumer connections (Gallant, et al. 2025: 1).

The desired state of a C-2-C e-commerce platform for these informal traders is a platform that has localised trust through secure inline payments and verified seller identities to prevent fraud. Moreover, the platform should also be designed with the local context in mind by supporting multiple languages as well as integration with local payment gateways and delivery logistics for rural areas.

1.2) Needs/problems

There are various issues that can negatively impact the integration of a C-2-C e-commerce platform in South Africa. These issues can include potential e-commerce users fearing that they could be deceived on the e-commerce platform from dishonest sellers, such as the seller not delivering the product that the user has paid for (Netshirando, V., et al. 2024: 546). Dishonest sellers on an e-commerce platform create an appearance of the platform being unreliable, which can drive potential and already existing users away from the concept of a C-2-C e-commerce platform.

Moreover, an online system like an e-commerce platform will accumulate vast amounts of sensitive data, which makes the platform a target for cyber threats (Mankolo Xaverine Lethoko 2024: 89). Therefore, potential sellers and buyers are reluctant to utilise an e-commerce platform where payments are performed digitally due to the possibility for their sensitive, financial data to be stolen from cybercriminals. Successful cyber security attacks on an e-commerce platform can greatly degrade user trust in the system and discourage potential users from using the system.

In addition, there is spatial inequality with South African infrastructure where informal traders in rural areas and townships can experience structural barriers like connectivity deficits (Suarez Clovis, O.N. 2025: 108). The lack of technological infrastructure in rural areas can discourage or entirely prevent individuals from utilising most online services including e-commerce platforms.

Furthermore, the delivery logistics for a C-2-C e-commerce platform for rural areas face various issues such as vehicle routing problems due to poor infrastructure, traffic congestion and tight delivery windows for home deliveries (*Electronic Commerce Research*. 2025: 423). Poor local delivery logistics along with other local aspects that aren't supported such as multiple languages and local payment methods can further discourage potential users from using an e-commerce platform.

1.3) Goals/Objectives

The objectives that address the problems identified for a C-2-C e-commerce platform include:

- Implement a mechanism to counteract dishonest vendors
 - Consumers and sellers should be required to follow an ethical agreement provided by an e-commerce platform to encourage transparency between both parties (Singh, N., et al. 2024: 2). Sellers agreeing to follow this agreement can enable the e-commerce platform to remove dishonest sellers who don't follow the agreement, which can discourage future dishonest sellers and increase the number of ethical vendors on the platform.
- Implement security mechanisms to protect online payment details
 - An e-wallet refers to a payment application that is linked to a user's bank account that enables users to make payments with methods like unique QR codes (Rahi, S., et al. 2024: 4819). Therefore, e-wallet functionality should be implemented into the e-commerce platform to ensure that online payments are secure, because the user will not have to provide their sensitive banking information when making payments.
- Improve data efficiency to combat connectivity issues
 - A Content Delivery Network (CDN) refers to geographically distributed system of servers situated at the edge of data centers that enables specific content from providers, like e-commerce and entertainment platforms, to be cached near end users (Gao, Z., et al. 2026: 2). Important data that is required for our e-commerce platform to function can be configured to be cached in CDNs to improve the e-commerce system's performance for users in rural areas with poor network infrastructure, especially for vendors who will
- Implement different delivery options to improve delivery logistics for varying areas
 - A similar e-commerce platform, Takealot, utilises convenient pickup points at various Pick n Pay stores throughout South Africa where customers can receive their Takealot orders when visiting the store (Disco Digital Media, Inc. 2025). Our C-2-C e-commerce platform can utilise a similar approach where the user can specify where they would like their order to be delivered. For rural areas with poor road infrastructure the user can have their orders delivered to a second location like a local store to reduce the impact of potential delivery challenges by shortening the delivery driver's travel distance.
- Implement support for local elements
 - When Amazon started to integrate itself into South Africa, it provided features such as onboarding tools, payment processing solutions and seller promotional features to assist to assist local vendors with their operations (Disco Digital Media, Inc. 2024). Our e-commerce platform can utilise these features with additional functionality like multiple language support to ease the adoption of online trading for new and already existing local enterprises by reducing potential knowledge barriers that could discourage them from using the platform.

1.4) Procedures/Scope of work

The frontend of the e-commerce platform will be developed with HyperText Markup Language (HTML) to define the structure, Cascading Style Sheets (CSS) that defines the layout and style as well as utilise JavaScript for client-side scripting due to its various libraries and frameworks. Additionally, the backend of the system will use the Hypertext Preprocessor (PHP) for server-side scripting with the MySQL database to store user data. Therefore, these development tools provide the necessary functionality to ensure that the e-commerce platform is able to perform all objectives that were identified.

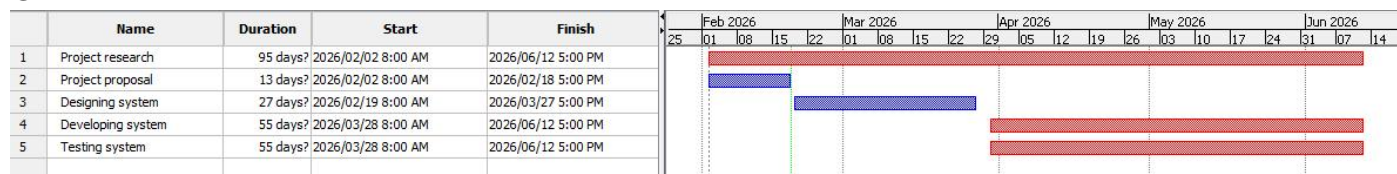
Moreover, user training should also be considered when developing the e-commerce platform, because users from rural areas are less likely to effectively utilise an e-commerce system in comparison to users from urban areas that have strong access to network infrastructure. Training for these end-users can be implemented through manuals that provide instructions on how to use the system as well as a help tool that provides instructions in a brief format (Oinas-Kukkonen, H., et al. 2010: 102).

Furthermore, thorough testing must be performed throughout the development of the e-commerce platform to ensure that unidentified, problematic issues are discovered and resolved before the system is available to users. Various testing tools can be utilised to test different aspects of the e-commerce platform including Load Runner to test the load balancing capability of the system, WAPT to stress test the system and Fortify to test the security of the system (Lakshmi, D.R. & Mallika, S.S. 2017: 2137-2138).

1.5) Timetable

	Description of work	Start and end dates
Deliverable 1	Create a project proposal for a C-2-C e-commerce platform.	02/02/2026 – 27/02/2026
Deliverable 2	Design and develop a C-2-C e-commerce system based on the project proposal.	28/02/2026 – 05/06/2026
Deliverable 3	Present the built C-2-C e-commerce platform.	06/06/2026 – 12/06/2026

Gantt chart:



1.6) Conclusion

There is a prominent shift in consumer behaviour towards online shopping due to convenience and it's likely that this attitude to e-commerce will continue to persist in South Africa (Katrodia, A. 2025: 632). However, local entrepreneurs and vendors are resistant to this change despite the possibility for expansion, because their lack of knowledge and small size act as barriers to adapting in the e-commerce industry (Madzvamuse, S., et al. 2024: 447).

Therefore, the proposed C-2-C e-commerce platform is designed to provide local entrepreneurs, especially in rural areas, with the necessary resources, infrastructure and training to adapt to the e-commerce industry. Joining this e-commerce platform can assist in the expansion of the informal economy by enabling entrepreneurs from rural areas to conduct business with consumers outside their community. The additional customers can allow these entrepreneurs to further develop their business to become more profitable, which can further expand the informal economy.

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